The *Connector* file relationships (server-side)

This section describes the *XML* and configuration files in the server-side *Connector* folder.

Connectors and *Providers*

A *Connector* is a specialised software component for managing the connection between your *Client* application and its database.

To use the terminology of *Object Orientation* – a *Connector* is (similar to) a *Class* – and a *Provider* is (similar to) an *Object* derived from that *Class* although it's an *Object* that is specialised for your *Client* application.

The simplest *Client* applications do not need the *application* or *script* files shown in the following illustration.

For information on the *application* and *script* files see the flyingSPARK document:

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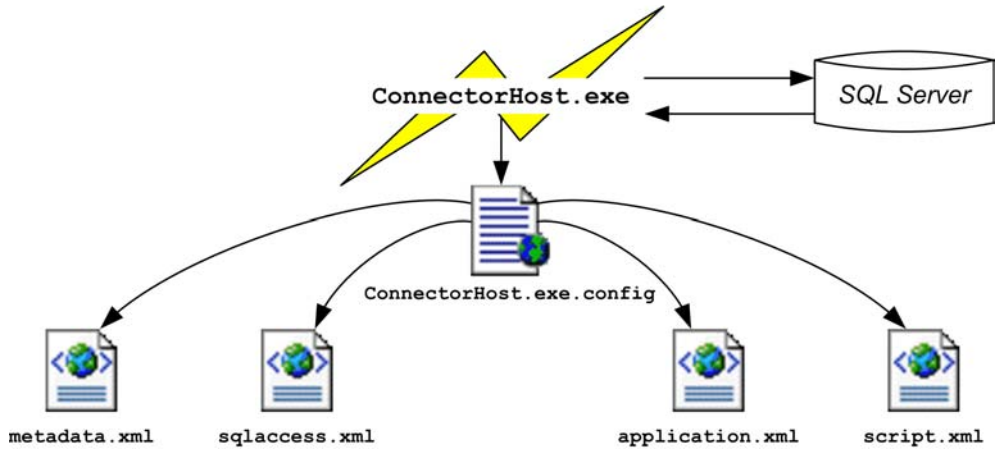


Exact copies of your *Client* application's *metadata* file must exist in the *Connector* folder (server-side) and the *Client* application folder (client-side).

The *ConnectorHost.exe.config* file

When the *ConnectorHost* runs it opens the *ConnectorHost.exe.config* file. The *ConnectorHost.exe.config* file defines the configuration of the *Provider* (a specific instance of a *Connector*).

It is likely that you'll have to customise your *ConnectorHost.exe.config* file for each *Client* application. The following extract shows the most significant extracts from the *ConnectorHost.exe.config* file used for *Sample Client*.

> The string values between and including the curly-brackets are non-literal *place-holders* that are explained below.

```
  …
  <add key="SystemDBConnection" value="server={SQLSERVER};
       user id={SQLUSER};password={SQLUSERPASSWORD};database=fsSystemKit;
       integrated Security=no;persist security info=no"/>
  …
<flyingSPARK>
  <SAMPLEProviderURI>
    <add key="ObjectName" value="SAMPLE" />
    <add key="MetaXmlFile" value="sample_metadata.xml" />
    <add key="SqlAccessFile" value="sample_sqlaccess.xml" />
    <add key="SqlCodeGenerator" value="flyingSPARK.SQLConnector.SQLServer2000.dll" />
    <add key="SqlClient" value="flyingSPARK.SQLConnector.SQLClient.dll" />
    <add key="ConnectionString"
         value="server={SQLSERVER};user id={SQLUSER}; password={SQLUSERPASSWORD};
         database=WIL_Sample;integrated security=no;persist security info=no" />
    <add key="debug" value="true" />
  </SAMPLEProviderURI>

  <SQLAuthURI>
    <add key="ObjectName" value="SQLAuth" />
    <add key="ConnectionString"
         value="server={SQLSERVER};user id={SQLUSER}; password={SQLUSERPASSWORD};
         database=WIL_Sample;integrated security=no;persist security info=no" />
    <add key="UsersTable" value="mobile_user" />
    <add key="UserName" value="username" />
    <add key="Password" value="password" />
  </SQLAuthURI>
</flyingSPARK>
```

> The strings that make up the *<SAMPLEProviderURI>* and *<SQLAuthURI>* nodes are arbitrary, but must be the same strings that appear as the value of the *name* attribute within the *section* nodes at the beginning of the *ConnectorHost.exe.config* file.

In the above extract (*<add key="SystemDBConnection" …* node):

- The string value *{SQLSERVER}* of the *Data Source* attribute is the name of a *SQL Server* (software server)
- The string value *{SQLUSER}* of the *User ID* attribute is the *Username* required for the system to access the database
- The string value *{SQLUSERPASSWORD}* of the *Password* attribute is the *Password* required for the system to access the database
- The string value *fsSystemKit* of the *database* attribute is the name of the database used by the *Wireless Intelligence Layer* components to store their data

In the above extract (*<add key="ConnectionString"* … nodes):

- The string value *WIL_Sample* of the *database* attribute is the name of the database
- The string value *{SQLSERVER}* of the *server* attribute is the name of the *SQL Server* (software server) that holds the database

> In this extract – the first *ConnectionString* node points at the database that contains the data used by *Sample Client*. The second *ConnectionString* node points at the database that contains the *Usernames* and *Passwords* of the *Users* of *Sample Client*.

- The string value *{SQLUSER}* of the *User ID* attribute is the *Username* required for the system to access the database
- The string value *{SQLUSERPASSWORD}* of the *Password* attribute is the *Password* required for the system to access the database

In the above extract (*<add key="UsersTable"* … node):

- The string value *mobile_user* of the *value* attribute is the name of the database table that holds the *Usernames* and *Passwords* of your *Client* application

In the above extract (*<add key="UserName"* … node):

- The string value *username* of the *value* attribute is the name of the column in *mobile_user* that holds the *Usernames* of *Sample Client's Users*

In the above extract (*<add key="Password"* … node):

- The string value *password* of the *value* attribute is the name of the column in *mobile_user* that holds the *Passwords* of *Sample Client's Users*

The *SAMPLEProviderURI* nodes

In the above extract (*<add key="ObjectName" value="SAMPLE" />* node):

- The string value *SAMPLE* of the *value* attribute is the name of the *Provider*, which must be reflected in:
  - ✓ The *metadata* file as the string value of the *name* attribute of the *provider* node
  - ✓ The *screens* file as the string value of the *provider* attribute of the *window* node

In the above extract (*<add key="MetaXmlFile" value="sample_metadata.xml" />* node):

- The string value *sample_metadata.xml* of the *value* attribute is arbitrary, but must be the filename of your *metadata* file reflected in:
  - ✓ The *metalist* file as the string value of the *definition* attribute of the *provider* node

In the above extract (*<add key="SqlAccessFile" value="sample_sqlaccess.xml" />* node):

- The string value *sample_sqlaccess.xml* of the *value* attribute is arbitrary, but must be the filename of your *sqlaccess* file

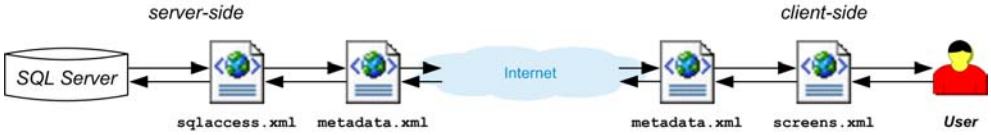
The *SQLAuthURI* nodes

In the above extract (*<add key="ObjectName" value="SQLAuth" />* node):

- The string value *SQLAuth* of the *value* attribute is arbitrary, but must be reflected in:
  - ✓ The *application* file as the string value of the *name* attribute of the *authenticator* node

The *sqlaccess* file

The *sqlaccess* file maps the *Entities* in the *metadata* file on to the database defined in the *Connection String* nodes of the *ConnectorHost.exe.config* file.



For details on the *sqlaccess* file see the flyingSPARK document:

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The *Core* file relationships (server-side)

This section describes the configuration files in the server-side *Core* folder.

The *svc* file

The following is an extract from a *svc* file (*svc.cfg*):

```
dynamic CDataHub Service_Object * datahub:_make_CDataHub () "netserver.cfg 36000 3900"
```

In the above extract:

- The string value *netserver.cfg* refers to the *netserver* file
- The string value *36000* is the *Port* of the *Core Server* used by the *Client* device. This number must and be reflected in the client-side *config* file (see The *config* file (page 7))
- The string value *3900* is the *Port* of the *Core Server* used by the *Server Manager*

Refers to:

The *svc* file refers to the *netserver* file (see below)

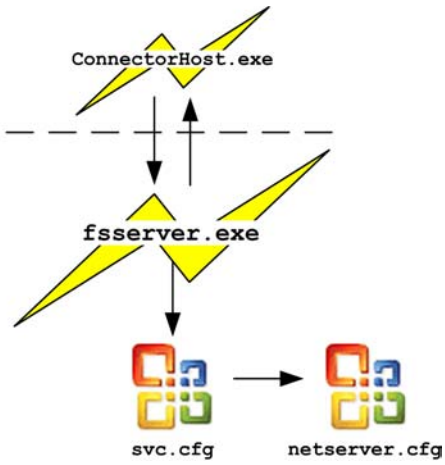
The *netserver* file

Although you can rename this file it is unlikely that you will need to. If you do rename it the new name must be reflected in the *svc* file described above.

A careful comparison of the *netserver* file and *ConnectorHost.exe.config* file will reveal the essential corresponding string values.

> The *Provider* used by your *Client* application is named in the *netserver* file.

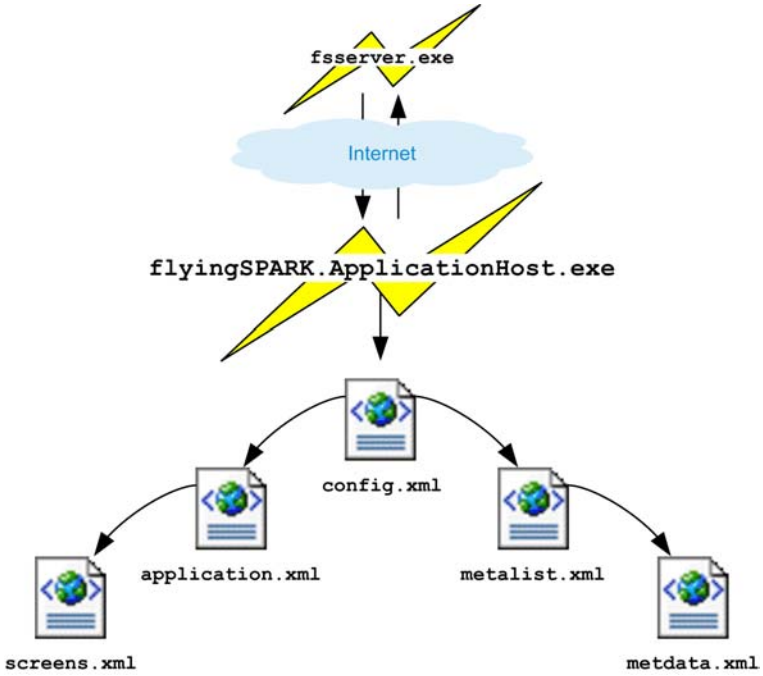
The following illustration shows the relationships between the *Core Server* (*fsserver.exe*) and its configuration files.



The *Client* file relationships (client-side)

From the point of view of your *Client* application's *User* (the screens they see) your *Client* applications are defined in the *screens* file.

Your *Client* applications are actually a *Package* of dependent *XML* files as shown in the following illustration.



When the *flyingSPARK.ApplicationHost.exe* (which you can rename) runs on the *Client* device it opens the *config* file – the top-level *XML* file.

The *config* file

The *config* file (*config.xml*) (which cannot be renamed) contains values including the *IP Address* and *Port Number* of your *Core Server* that your *Client* application will connect to when it's logged on.

This port number is reflected in the server-side *svc* file (*svc.cfg*) (see The *svc* file (page 6)).

Refers to:

The *metalist* file

Referred to between the *metadata* nodes:

```
<metadata>sample_metalist.xml</metadata>
```

The *application* file

Referred to between the *ui* nodes:

```
<ui>sample_application.xml</ui>
```